

DENTAL MATERIALS

LABORATORY ACTIVITY 4



Name _____ Date _____
Course _____

LABORATORY ACTIVITY 4 – IMPRESSION MATERIALS

Impression Compound

Learning Outcomes:

1. Classify the different types of impression materials and its composition.
2. Discuss the physical and mechanical properties of impression and evaluate how these affect their clinical performance.
3. Demonstrate correct manipulation techniques for different types of impression materials while identifying factors that affect impression accuracy.



Impression Compound



LABORATORY ACTIVITY 4: IMPRESSION COMPOUND

UNIT 1 –IMPRESSION COMPOUND

UNIT LEARNING OUTCOMES:

1. To study the general properties of an impression compound



MATERIALS:

Stock tray (upper and lower)

Water bath

Modeling plastic/Compound

Compound Heater

Carving knife

Edentulous cast

Sand paper

Prosthodontic scissor and carvers

Basin

Petroleum Jelly

PROCEDURE:

1. Allow the modelling plastic to soften in hot water
2. Knead and roll the compound with your fingers
3. Note and record the condition of the compound
4. Soften again in hot water, roll, and place it into the stock tray
5. Fix the soft modelling compound with the fingers
6. Take an impression of the edentulous cast
7. Allow the impression compound to harden completely in the edentulous cast.
8. Remove from the edentulous cast when hard
9. Trim the excess on the impression with a sharp knife by following the correct outline of the impression

explain

1. Which property of impression compound allows it to be softened and reused during manipulation?
2. Why is impression compound unsuitable for recording deep undercuts?
3. What is the purpose of kneading impression compound after heating?
4. A student heats impression compound but notices that some portions remain hard while others are very soft. What should the student do?
5. A dentist attempts to remove an impression compound impression from an area with a pronounced undercut. The impression becomes distorted. What property of impression compound explains this problem?

Laboratory Activity 4

Impression Compound

Completed responses | Follows source worksheet page 4

1. Which property of impression compound allows it to be softened and reused during manipulation?

Answer: Its thermoplastic and reversible physical behavior. It softens when heated and hardens again on cooling without a chemical setting reaction, so it can be reheated and remolded for the same patient.

2. Why is impression compound unsuitable for recording deep undercuts?

Answer: After cooling it is rigid and has very low elasticity. It cannot flex safely around a deep undercut, so removal may distort the impression, fracture it, or injure tissues.

3. What is the purpose of kneading impression compound after heating?

Answer: Kneading distributes heat throughout the mass, removes hard lumps, and produces uniform softness, plasticity, and consistency before the material is loaded into the tray.

4. Some portions remain hard while others are very soft. What should the student do?

Answer: Return the material to a controlled water bath, soften it uniformly, and knead it again until the whole mass has an even consistency. Avoid direct flame or excessive heat, which can over-soften or damage the material.

5. Why did an impression distort when removed from a pronounced undercut?

Answer: The set compound is rigid and has poor elastic recovery. Because it cannot flex over the undercut and return to its original shape, removal produces permanent deformation.

DENTAL MATERIALS

LABORATORY ACTIVITY 5

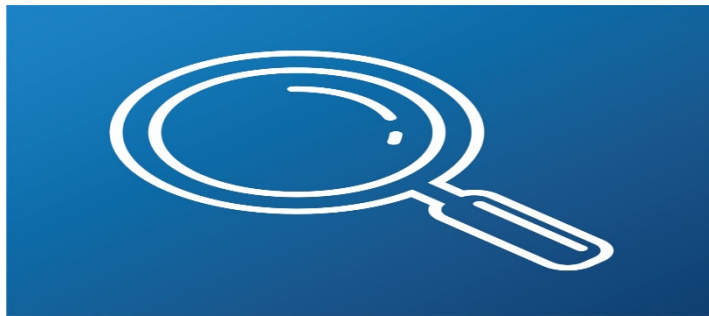


Name _____ Date _____
Course _____

LABORATORY ACTIVITY 5 – Preliminary Cast Construction: Inverted Technique and Boxing Method

Learning Outcomes:

1. Explain the importance of accelerators and retarders during cast construction
2. Differentiate between various types of dental waxes.
3. Demonstrate the different method used during cast construction
4. To construct a model in which the denture will be developed



LABORATORY ACTIVITY 5: PRELIMINARY CAST CONSTRUCTION

UNIT 1 –PRELIMINARY CAST CONSTRUCTION

UNIT LEARNING OUTCOMES:

1. Demonstrate the correct handling and manipulation techniques of Plaster of Paris for specific dental procedures and laboratory processes.
2. Outline the steps in the cast production procedure



Instruments:

- Plaster bowl and spatula
- Alcohol lamp with denature alcohol
- Wax spatula
- Sharp prosthodontic knife or a pair of scissors
- Vibrator
- Model trimmer

Materials:

- Plaster of Paris
- Utility wax (rope wax)
- Base plate wax
- Boxing wax
- Modelling clay

Procedure:

1. Prepare the impression for casting. It should be washed to remove saliva and must be disinfected. Areas that are not important may be trimmed with a pair of scissors or a sharp knife. Allow the impression to dry; paper towels may be used for this purpose.
2. Box the impression. Stick utility wax around impression. This is done by positioning the wax rope about 3-4 mm below the peripheral roll of the impression
 1. The maxillary impression
 - The heading of the wax should not cover any tissue part of the impression at the posterior palatal seal area. Place the wax beyond this area.
 2. The mandibular impression
 - A piece of base plate wax is cut and shaped to fit the lingual area of the impression (use modelling clay if baseplate wax is not available). This is approximated along the wax beading area.



Figure 15. The inversion method of making the base of the cast

3. Using a heated wax spatula, seal the beading wax along the periphery of the impression, including the base plate wax at the lingual aspect of the mandibular impression.

4. Dispense enough Plaster of Paris for the cast and the base. Mix the correct water-powder ratio (1:3 w/v) into a homogenous mixture. Pour one impression at a time. Do not forget to pour Plaster of Paris onto the lingual aspect of the mandibular impression. Allow the stone to partially set.
5. Put Plaster of Paris intended for the base onto a flat surface (8" x 8" ceramic tile) and create a circular mound. Invert the poured impression with partially set stone onto this mound (Fig. 15). Work on the side of the mound with a spatula. Create a base by shaping the mound to conform to the area of the wax bead. Allow the stone to completely set for about 30 minutes. (INVERTED TECHNIQUE)
6. . Retrieve the casts from the impressions. Examine them for any void spaces or nodules.
 - Major defects in cast production may necessitate re-doing the entire impression, as these may affect the shape of the denture base.
7. With the use of a model trimmer, smoothen the sides of the casts, but do not over-trim.
 - Use the thickness of the beading wax area as a guide.
 - The model must not be trimmed beyond this, as it will be used in preparing the land area of the casts.
8. With the use of a sharp knife, level the land area of the cast. The height of the land area depends on your ability to finger-access the mucobuccal fold reflection in the cast.

explain

1. What happens when Plaster of Paris is mixed with water?
2. Why should the manufacturer's recommended water-to-powder ratio be followed?
3. Why is vibration used when pouring Plaster of Paris into an impression?
4. A student prepares Plaster of Paris using excessive water. After setting, the cast is soft and weak. What is the most likely explanation?
5. A dentist wants to produce a strong and accurate gypsum cast. Which procedure is most appropriate?
6. What factors can influence the initial and final setting time of a gypsum product? How can accelerators and retarders be used

Laboratory Activity 5

Preliminary Cast Construction - Plaster of Paris

Completed responses | Follows source worksheet page 8

1. What happens when Plaster of Paris is mixed with water?

Answer: Calcium sulfate hemihydrate dissolves and reacts with water to form calcium sulfate dihydrate. Dihydrate crystals precipitate and interlock, releasing heat and changing the fluid mix into a rigid solid.

2. Why should the manufacturer's recommended water-to-powder ratio be followed?

Answer: The specified ratio gives proper flow, setting behavior, strength, surface hardness, detail reproduction, and dimensional accuracy. Excess water creates porosity and weakness; too little water makes the mix thick and difficult to adapt accurately.

3. Why is vibration used when pouring Plaster of Paris into an impression?

Answer: Gentle vibration helps the slurry flow into small details and allows trapped air to escape, reducing bubbles and surface voids. Excessive vibration should be avoided because it can encourage segregation or introduce defects.

4. Why is a cast made with excessive water soft and weak after setting?

Answer: Only part of the added water is consumed by the setting reaction. The excess later evaporates and leaves capillary pores between crystals, reducing density, hardness, and compressive strength.

5. Which procedure is most appropriate for producing a strong, accurate gypsum cast?

Answer: Measure water and powder exactly as directed; add powder to water; allow wetting; spatulate for the stated time to a smooth mix; pour in small increments under gentle vibration; and leave the cast undisturbed until it has fully set before separation.

6. What influences gypsum setting time, and how are accelerators and retarders used?

Answer: Setting time is affected by the water-powder ratio, water temperature, mixing time and speed, particle size, storage or contamination, and additives. More vigorous or prolonged spatulation usually shortens setting by creating more crystal nuclei. An accelerator such as potassium sulfate shortens the set; a retarder such as borax or citrate lengthens it. Use only controlled, manufacturer-approved amounts because additives can also alter expansion and strength.

DENTAL MATERIALS

LABORATORY ACTIVITY 6

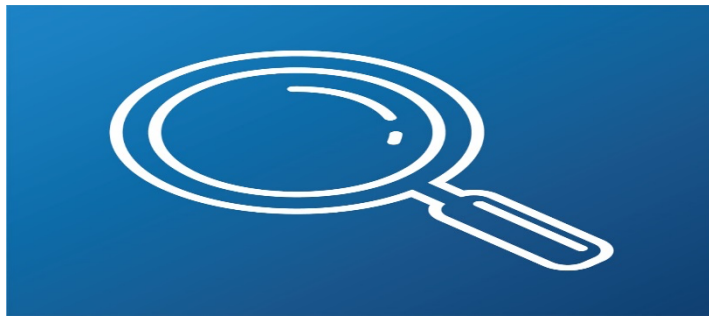


Name _____ Date _____
Course _____

LABORATORY ACTIVITY 6 – Final Impression

Learning Outcomes:

1. Explain the importance of water-powder ratio during manipulation of the impression material
2. Differentiate between various types of impression materials
3. Demonstrate how to make a good impression
4. To know how to make an accurate impression



LABORATORY ACTIVITY 6: Final Impression

UNIT 1 –FINAL IMPRESSION

UNIT LEARNING OUTCOMES:

1. Demonstrate the correct handling and manipulation techniques mixing impression materials.
2. Outline the steps in the final impression procedure



Materials needed:

Mixing bowl	Spatula
Perforated Tray	Boxing Wax
Preliminary Cast	Alginate
Modelling Clay	

- Making an accurate dental impression using alginate requires precise measurement, fast mixing, and proper tray placement.

PROCEDURE:

A. Preparation and Tray Selection

- a. Select the correct size of maxillary (upper) or mandibular (lower) impression tray so it covers all teeth comfortably with a few millimeters of clearance
- b. Fluff the alginate powder by gently turning the closed container upside down to aerate the material
- c. Measure the exact ratio of room-temperature water and powder recommended by the manufacturer. Cooler water slows setting time, while warmer water speeds it up

B. Mixing the Alginate

- a. Pour the measured water into a flexible mixing bowl first, then add the powder to minimize air bubbles.
- b. Use a wide spatula to wet the powder, then vigorously press and wipe the mixture flat against the inner walls of the bowl.
- c. Mix continuously for about 30 to 45 seconds until the paste is smooth, creamy, and uniform without lumps.

C. Loading the Tray

- a. Scoop the creamy alginate with the spatula and load it into the impression tray.
- b. Shape the material evenly, placing slightly more in the front (anterior) section.
- c. Smooth the surface with a clean, wet finger to remove surface imperfections. Wipe a small amount of excess mix directly onto the patient's occlusal (chewing) surfaces of the teeth to prevent trapped air pockets

D. Seating the Tray (Applied for live patient)

- a. Retract the lips and cheeks, then insert the tray into the mouth from the side or angle.
- b. Seat the back (posterior) teeth first, then rotate or push down the front part.
- c. Apply firm, steady, and equal pressure on both sides until a small amount of alginate edges out of the borders. Hold the tray completely still while the patient breathes through their nose

E. Setting and Removal

- a. Wait for the material to set completely, typically 2 to 3 minutes (check your specific product label and test the leftover material in the bowl with a fingernail)
- b. Break the suction seal by placing a finger in the back corner of the tray and pushing down firmly, or ask the patient to puff air against the sealed edges
- c. Remove the tray quickly in one firm motion, rinse the impression with water, inspect it for details, and disinfect it immediately

explain

1. Which type of impression material is alginate classified as? Why?
2. What property of alginate allows it to reproduce the general anatomy of the oral cavity? Why?
3. Why should an alginate impression be poured as soon as possible after removal from the mouth?
4. What happens when alginate loses water to the environment? Why?
5. What factor primarily controls the working and setting time of alginate? Explain
6. A student adds excessive water to alginate to make the material easier to mix. What is the most likely effect? Why?
7. Why should the impression tray be properly selected before alginate manipulation?
8. A dentist obtains an alginate impression with poor surface detail and weak areas. Analysis reveals that the operator used excessive water and failed to follow the recommended mixing procedure. Explain briefly the cause of the problem.

Laboratory Activity 6

Final Impression - Alginate

Completed responses | Follows source worksheet page 12

1. Which type of impression material is alginate classified as, and why?

Answer: Alginate is an elastic, irreversible hydrocolloid. It changes chemically from a sol to a calcium-alginate gel and cannot be returned to the sol state by reheating.

2. What property lets alginate reproduce the general anatomy of the oral cavity?

Answer: Before gelation, its hydrophilic, flowable sol wets oral tissues and adapts to their contours. After setting, its elasticity allows the impression to pass over moderate undercuts with limited permanent deformation.

3. Why should an alginate impression be poured as soon as possible?

Answer: Alginate is dimensionally unstable. Water loss by evaporation or syneresis causes shrinkage, while water uptake by imbibition causes expansion; either change makes the resulting cast inaccurate.

4. What happens when alginate loses water to the environment, and why?

Answer: It shrinks and distorts. Water leaves the gel by evaporation and syneresis while the gel network contracts, reducing the impression's dimensions.

5. What primarily controls the working and setting time of alginate?

Answer: The main operator-controlled factor is water temperature: warmer water speeds the reaction and shortens working and setting time, while cooler water lengthens them. The material's formulation sets the baseline, so the manufacturer-specified ratio should not be changed to control time.

6. What is the likely effect of adding excessive water, and why?

Answer: The mix becomes too fluid, usually sets more slowly, and forms a weaker gel with reduced tear strength and poorer detail. The excess water dilutes the reacting ingredients and leaves a less dense structure.

7. Why should the impression tray be properly selected before manipulation?

Answer: A correctly sized tray covers all required anatomy, provides a uniform bulk of material, supports the gel, and permits stable seating and removal. A poor fit can cause missed areas, thin sections, tearing, movement, or distortion.

8. Why did excessive water and improper mixing produce poor detail and weak areas?

Answer: Excess water diluted the mix and reduced gel strength, viscosity, and tear resistance. Inadequate spatulation left powder incompletely wetted and trapped lumps or air, producing a nonuniform surface with voids, weak zones, and inaccurate detail.

DENTAL MATERIALS

LABORATORY ACTIVITY 7

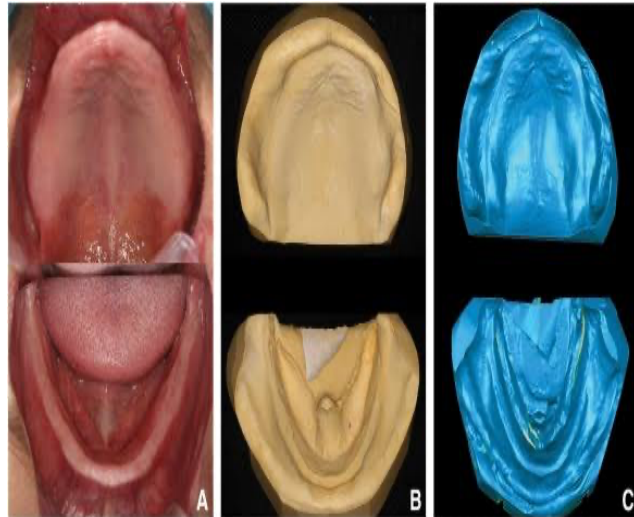


Name _____ Date _____
Course _____

LABORATORY ACTIVITY 7 – Master Cast Construction: Inverted Technique and Boxing Method

Learning Outcomes:

1. Explain the importance of accelerators and retarders during cast construction
2. Differentiate between various types of gypsum products such as dental stone and Plaster of Paris.
3. Demonstrate the different method used during cast construction
4. To construct a model in which the denture will be developed



LABORATORY ACTIVITY 7: MASTER CAST CONSTRUCTION

UNIT 1 –MASTER CAST CONSTRUCTION

UNIT LEARNING OUTCOMES:

1. Demonstrate the correct handling and manipulation techniques of dental stone for specific dental procedures and laboratory processes.
2. Outline the steps in the cast production procedure



Instruments:

Plaster bowl and spatula
Alcohol lamp with denature alcohol
Wax spatula
Sharp prosthodontic knife or a pair of scissors
Vibrator
Model trimmer

Materials:

Dental stone
Utility wax (rope wax)
Base plate wax
Boxing wax
Modelling clay

Procedure:

1. Prepare the impression for casting. It should be washed to remove saliva and must be disinfected. Areas that are not important may be trimmed with a pair of scissors or a sharp knife. Allow the impression to dry; paper towels may be used for this purpose.
2. Box the impression. Stick utility wax around impression. This is done by positioning the wax rope about 3-4 mm below the peripheral roll of the impression
3. The maxillary impression
 - The heading of the wax should not cover any tissue part of the impression at the posterior palatal seal area. Place the wax beyond this area.
4. The mandibular impression
 - A piece of base plate wax is cut and shaped to fit the lingual area of the impression (use modelling clay if baseplate wax is not available). This is approximated along the wax beading area.



Figure 15. The inversion method of making the base of the cast

3. Using a heated wax spatula, seal the beading wax along the periphery of the impression, including the base plate wax at the lingual aspect of the mandibular impression.

4. Dispense enough dental stone for the cast and the base. Mix the correct water-powder ratio into a homogenous mixture. Pour one impression at a time. Do not forget to pour dental stone onto the lingual aspect of the mandibular impression. Allow the stone to partially set.
5. Put dental stone intended for the base onto a flat surface (8" x 8" ceramic tile) and create a circular mound. Invert the poured impression with partially set stone onto this mound (Fig. 15). Work on the side of the mound with a spatula. Create a base by shaping the mound to conform to the area of the wax bead. Allow the stone to completely set for about 30 minutes.
6. . Retrieve the casts from the impressions. Examine them for any void spaces or nodules.
 - Major defects in cast production may necessitate re-doing the entire impression, as these may affect the shape of the denture base.
7. With the use of a model trimmer, smoothen the sides of the casts, but do not over-trim.
 - Use the thickness of the beading wax area as a guide. The model must not be trimmed beyond this, as it will be used in preparing the land area of the casts.
8. With the use of a sharp knife, level the land area of the cast. The height of the land area depends on your ability to finger-access the mucobuccal fold reflection in the cast.

explain

1. Why is dental stone generally stronger than Plaster of Paris?
2. What is the likely effect of excessive spatulation on the setting time of dental stone?
3. A dental student prepares a dental stone cast using too much water. After setting, the cast is weak and easily abraded. What is the most likely cause?
4. A dental stone cast contains numerous small voids on its surface. Which manipulation error is most likely responsible?
5. Which characteristic of dental stone is important for producing an accurate working cast?

Laboratory Activity 7

Master Cast Construction - Dental Stone

Completed responses | Follows source worksheet page 17

1. Why is dental stone generally stronger than Plaster of Paris?

Answer: Dental stone is mainly alpha-calcium sulfate hemihydrate, whose particles are denser, more regular, and less porous than the beta-hemihydrate particles of plaster. It therefore requires less mixing water and sets with less residual porosity and greater strength.

2. What is the likely effect of excessive spatulation on setting time?

Answer: It usually shortens the setting time because continued mixing creates more crystal nuclei and speeds crystal growth. It can also increase setting expansion and may incorporate air if performed improperly.

3. Why is a dental stone cast made with too much water weak and easily abraded?

Answer: The excessive water increases the water-powder ratio. Water not used in hydration later evaporates, leaving pores and widely spaced crystals that lower surface hardness, abrasion resistance, and overall strength.

4. Which manipulation error most likely caused numerous small surface voids?

Answer: Air was trapped because the stone was poured too quickly or without gentle vibration. Adding small increments from one area while vibrating lightly lets the advancing slurry displace air from fine details.

5. Which dental-stone characteristic is important for an accurate working cast?

Answer: High dimensional accuracy - especially low, controlled setting expansion - is essential. Fine-detail reproduction, adequate strength, and abrasion resistance are also needed so the cast remains accurate during laboratory procedures.